

PROBLEM STATEMENT #63 | Sustainability & Green Tech

Community Solar Credit Allocation Manager

1. Problem Context & Overview

A clean energy sharing portal ingesting rooftop solar generation metrics, facilitating peer-to-peer solar credit transfers within a neighborhood microgrid, and tracking monthly utility billing offsets.

Target Stakeholders / Actors: *Prosumer Resident, Co-op Manager*

2. Sample Functional Requirement (FR) Guideline

Sample Requirement: FR-001 [Priority: High]

Description: The system shall record solar generation units (kWh) from smart meter feeds and allow prosumers to transfer surplus credits to neighbor accounts.

Sample Acceptance Criteria: Pass: Transferred credits deduct from prosumer balance and apply to neighbor billing statement; Fail: Transfer exceeding balance.

3. Sample Non-Functional Requirement (NFR) Guideline

Sample Requirement: NFR-001 [Type: Performance & Security]

Description: The smart meter telemetry processing pipeline must validate and store hourly generation data for 500 households with 99.9% uptime.

Sample Acceptance Criteria: Pass: Benchmarking tests confirm target latency and security standards under simulated peak load.

Deliverables to Submit: [All through a GitHub Repo in your account]

- **Complete Requirements Table:**

- >> Exactly 5 Functional Requirements (FR-001 to FR-005)
- >> 2 Non-Functional Requirements (NFR-001 & NFR-002) with ID, Type, Description, Priority, Acceptance Criteria, and Rationale.

- **UML Use-Case Diagram:**

- >> Model all actors, primary use cases, and include at least one «include» and one «extend» relationship.

- **Use-Case Flow Specification:**

- >> 1-page document for one core use case detailing Preconditions, Postconditions, Main Success Scenario
- >> One Alternate Flow.